MODULE 01

Cybersecurity & Networking Foundations

Day 1 – Day 5 | Beginner Level

Learning Notes + Practical Lab Record

Module outcomes

- Explain cybersecurity fundamentals, the CIA Triad, common threats and attack surfaces.
- Understand networking basics including IP and MAC addresses, ports, protocols, OSI and TCP/IP models.
- Describe HTTP/HTTPS, DNS, DHCP, and authentication versus authorization.
- Set up a safe, isolated learning lab and inspect network information using permitted tools.
- Record observations and prepare a Cybersecurity Fundamentals Report.

Prepared as a beginner-friendly study document. Use all practical exercises only on your own device, a lab network, or systems for which you have explicit permission.

How to use these notes

This module is arranged as a five-day learning path. Read the concept notes first, complete the short activities, and write your own observations in the lab-record sections. Commands shown are for viewing or testing your own system and local lab; do not scan or test networks without permission.

Five-day study plan

Day	Focus	Expected output
1	Cybersecurity foundations, CIA Triad, threats and attack surfaces	Threat-and-risk notes
2	Networking basics: IP, MAC, ports and protocols	Local network information record
3	OSI/TCP-IP models and core protocols	Protocol comparison table
4	Authentication, authorization and safe lab setup	Lab setup checklist
5	Network observation, documentation and review	Final report and revision answers

Key terms

- **Asset:** something valuable that needs protection, such as data, a device or an account.
- **Threat:** a possible cause of harm, such as malware, phishing or a malicious insider.
- **Vulnerability:** a weakness that a threat could exploit.
- **Risk:** the possibility and impact of a threat exploiting a vulnerability.
- **Control:** a safeguard that reduces likelihood or impact, such as MFA, backups or encryption.

A simple risk model: Risk is commonly assessed by considering the likelihood of an event and the impact if it occurs. A high-impact event may deserve attention even when its likelihood is relatively low.

DAY 1 — Cybersecurity fundamentals

1.1 What is cybersecurity?

Cybersecurity is the practice of protecting computers, networks, applications and information from unauthorized access, misuse, disruption, modification or destruction. It combines people, processes and technology. Examples include using strong authentication, installing security updates, limiting user permissions, monitoring systems and maintaining backups.

1.2 The CIA Triad

Principle	Meaning	Example control
Confidentiality	Information is accessible only to authorized people or systems.	Encryption, access controls, least privilege.
Integrity	Information remains accurate and is changed only in authorized ways.	Hashes, digital signatures, change control.
Availability	Systems and information are accessible when needed.	Backups, redundancy, patching, recovery plans.

Example: If an attacker reads a private student database, confidentiality is affected. If marks are changed without permission, integrity is affected. If a college portal is unavailable during registration, availability is affected.

1.3 Common cybersecurity threats

- **Malware:** malicious software such as viruses, worms, trojans, spyware and ransomware.
- **Phishing:** deceptive messages or websites designed to trick people into revealing information or opening harmful files.
- **Password attacks:** attempts to guess, reuse or steal passwords. Password reuse makes several accounts vulnerable at once.
- **Denial of Service (DoS):** an attempt to make a service unavailable by exhausting its resources. Distributed DoS uses many sources.
- **Man-in-the-Middle (MitM):** an attacker positions themselves between communicating parties to observe or alter traffic.
- **Insider threat:** risk caused by someone with legitimate access, intentionally or accidentally.

1.4 Attack surface

An attack surface is the collection of points where an attacker could try to enter, affect or extract data from a system. It includes internet-facing services, open ports, user accounts, applications, APIs, wireless access, removable media, cloud storage and people. Reduce it by disabling unnecessary services, applying updates, removing unused accounts, using MFA and limiting exposure.

1.5 Basic security practices

- Use unique passwords stored in a reputable password manager; enable multi-factor authentication (MFA).
- Install operating-system and application updates from trusted sources.
- Back up important data and periodically test restoration.
- Be cautious with unexpected links, attachments and requests for credentials.
- Lock devices, use screen timeouts and secure Wi-Fi with modern encryption.
- Give each account only the permissions required for its task.

Day 1 quick activity: Choose a familiar service (email, online banking or a college portal). Identify one asset, one possible threat, one vulnerability and one control. Do not attempt to test or access another person's account.

DAY 2 — Networking fundamentals

2.1 What is a computer network?

A computer network is a group of connected devices that exchange data and share resources using agreed rules called protocols. Networks may be classified by coverage: PAN (personal), LAN (local), MAN (metropolitan) and WAN (wide area). The internet is a global network of interconnected networks.

2.2 IP address

An Internet Protocol (IP) address is a logical address used to identify an interface on an IP network and help route packets. IPv4 uses 32 bits, commonly written as four decimal octets, for example 192.168.1.10. IPv6 uses 128 bits and is written in hexadecimal groups, for example 2001:db8::10. The 2001:db8::/32 range is reserved for documentation examples.

- A private IPv4 address is used inside private networks and is not directly routed across the public internet.
- A public IP address is globally routable, subject to network and provider configuration.
- A static address is configured to remain fixed; a dynamic address is assigned for a period, commonly by DHCP.
- A subnet mask or prefix length identifies the network portion of an IP address.

2.3 MAC address

A Media Access Control (MAC) address is a link-layer identifier associated with a network interface. Ethernet MAC addresses are commonly 48 bits and shown as six hexadecimal pairs, for example 00:1A:2B:3C:4D:5E. A MAC address is used for delivery on a local link; routers use IP addressing to forward traffic between networks. MAC addresses can be randomized or changed by software, so they should not be treated as reliable proof of a device's identity.

2.4 IP address versus MAC address

IP address	MAC address
Logical network-layer address	Link-layer hardware/interface identifier
Used for routing between networks	Used for delivery on a local network segment
May change when a device joins another network	Often associated with an interface; may be randomized or changed

2.5 Ports and protocols

A port number identifies an application/service endpoint on a host. TCP and UDP use port numbers from 0 to 65535. A protocol defines communication rules, including message format and behavior. A port being listed as common does not mean it is open on every device.

Port	Common service	Purpose
20/21 TCP	FTP	Legacy file transfer; not encrypted by default.
22 TCP	SSH	Secure remote login and administration.
25 TCP	SMTP	Mail transfer between mail servers.
53 TCP/UDP	DNS	Domain name resolution.
67/68 UDP	DHCP	Automatic IPv4 configuration.
80 TCP	HTTP	Web traffic without transport encryption.
110 TCP	POP3	Mail retrieval; secure variants are preferred.

Port	Common service	Purpose
143 TCP	IMAP	Mail access; secure variants are preferred.
443 TCP	HTTPS	HTTP protected by TLS.
3389 TCP	RDP	Remote desktop service; restrict and secure carefully.

Day 2 quick activity: View the IP configuration on your own computer. Record the active interface, IPv4/IPv6 address (if shown), default gateway and DNS server. Do not publish these details publicly.

DAY 3 — Models and core protocols

3.1 OSI reference model

The Open Systems Interconnection (OSI) model divides network communication into seven conceptual layers. It helps explain where protocols and network functions operate and supports troubleshooting.

Layer	Name	Main responsibility	Example
7	Application	Services used by applications	HTTP, DNS, SMTP
6	Presentation	Data format, encoding, encryption concepts	Character encoding, TLS functions
5	Session	Establishes/manages communication sessions	Session management
4	Transport	End-to-end delivery, reliability and ports	TCP, UDP
3	Network	Logical addressing and routing	IP, routers
2	Data Link	Frames and local-link delivery	Ethernet, Wi-Fi, switches
1	Physical	Signals and physical transmission	Copper, fibre, radio

Memory aid (top to bottom): All People Seem To Need Data Processing. The OSI model is a teaching and troubleshooting model; real protocols do not always fit perfectly into one layer.

3.2 TCP/IP model

The TCP/IP model describes the architecture used by the internet. A common four-layer version is:

- **Application:** application protocols such as HTTP, DNS, SMTP and DHCP.
- **Transport:** TCP or UDP communication between processes using ports.
- **Internet:** IP addressing and packet routing.
- **Link / Network Access:** local network delivery over Ethernet, Wi-Fi or another link.

OSI and TCP/IP comparison

OSI model	TCP/IP model (four-layer view)
Application + Presentation + Session	Application
Transport	Transport
Network	Internet
Data Link + Physical	Link / Network Access

3.3 Core protocols

HTTP and HTTPS: HTTP is an application protocol used to transfer web resources. HTTPS is HTTP protected using TLS, which helps provide encryption, server authentication and message integrity in transit. Users should check for the correct domain name; the lock icon alone does not prove a site is trustworthy.

DNS: The Domain Name System translates names such as example.com into records including IP addresses. DNS commonly uses UDP port 53 for ordinary queries, while TCP is also used in several cases, including larger responses and zone transfers. Encrypted DNS options include DoT and DoH.

DHCP: Dynamic Host Configuration Protocol automatically provides network settings such as an IP address, subnet mask/prefix, default gateway and DNS servers. The familiar IPv4 sequence is Discover, Offer, Request, Acknowledge (DORA).

TCP: Connection-oriented transport that provides ordered, reliable delivery with retransmission and flow/congestion control. **UDP:** Connectionless transport with lower protocol overhead; it does not itself guarantee delivery or ordering.

Day 3 quick activity: For a website you visit, identify the domain name and explain (in your own words) how DNS and HTTPS contribute to loading it. Do not intercept other users' traffic.

DAY 4 — Identity, access and safe lab setup

4.1 Authentication versus authorization

Authentication answers “Who are you?” It verifies an identity using one or more factors. **Authorization** answers “What are you allowed to do?” It determines permissions after identity has been established.

Concept	Explanation	Example
Authentication	Verifies identity.	Signing in using a password and authenticator app.
Authorization	Checks allowed actions or resources.	A student can view marks but cannot edit them.
Accounting / Auditing	Records activity for accountability.	A log records sign-in time and administrative changes.

4.2 Authentication factors

- **Something you know:** password or PIN.
- **Something you have:** security key, smart card or authenticator device.
- **Something you are:** biometric characteristic such as fingerprint or face recognition.

Multi-factor authentication uses two or more different factor categories. Two passwords are not two-factor authentication because both are something you know.

4.3 Access-control principles

- **Least privilege:** grant only the minimum permissions needed.
- **Need to know:** provide access only when required for a task.
- **Role-based access control (RBAC):** assign permissions to roles, then assign users to roles.
- **Zero trust:** do not automatically trust a user or device because it is inside a network; verify access explicitly and continuously where appropriate.

4.4 Safe learning lab

A safe lab is an isolated environment where you can learn without affecting real users or production systems. Beginner options include your own computer, a virtual machine configured for host-only networking, or a dedicated practice environment that explicitly authorizes testing.

1. Use a computer and network you own or have written permission to use.
2. Install updates and use a standard (non-administrator) account for ordinary work.
3. If using virtual machines, choose host-only or an isolated internal network for exercises that do not require internet access.
4. Take a snapshot before experiments so you can restore the virtual machine.
5. Use sample data; never place real passwords, private documents or personal records in a practice lab.
6. Keep lab tools and test targets within the authorized environment.

Safety boundary: Do not scan public IP ranges, college networks, neighbours' Wi-Fi, or websites without explicit authorization. A tool being available does not make every target permissible.

DAY 5 — Network observation and documentation

5.1 Inspecting your own network configuration

The following commands display local network information. Command output differs by operating system. Review the output locally and redact public addresses, usernames, device names and other identifying information before sharing screenshots.

Operating system	Command	What it shows
Windows	ipconfig /all	IP configuration, gateways, DNS and adapter details.
Windows	getmac /v	MAC addresses associated with local adapters.
Windows	netstat -ano	Local connections/listening endpoints and process IDs.
macOS / Linux	ifconfig (if installed)	Network interface configuration; may be replaced by ip.
Linux	ip addr	IP addresses and interface state.
Linux	ip route	Routing table and default route.
macOS / Linux	ss -tuln (Linux); netstat -an (macOS)	Listening TCP/UDP sockets or network endpoints.

5.2 Understanding command output

- An interface is a network connection point, such as Wi-Fi or Ethernet.
- A default gateway is the router a device typically uses to reach destinations outside its local subnet.
- A DNS server resolves domain names into records.
- A listening socket indicates that a local program is waiting for network communication; it does not by itself prove a vulnerability.
- A process ID (PID) helps identify which local process owns a connection on supported systems.

5.3 Basic troubleshooting sequence

1. Check whether Wi-Fi or Ethernet is connected.
2. Check that the device has a valid IP address and subnet configuration.
3. Check the default gateway and whether the local router is reachable.
4. Check DNS resolution using a permitted test such as nslookup example.com.
5. Compare whether the issue affects one application, one device or multiple devices.
6. Record the time, symptom, test and result before changing settings.

5.4 Practical lab record

Fill this in using your own device. Do not include passwords, private keys or sensitive personal data.

Field	Your observation
Date and time	_____
Device / OS	_____
Network type (Wi-Fi/Ethernet)	_____

Interface name	_____
Private IP / prefix (redact before sharing)	_____
Default gateway (redact before sharing)	_____
DNS server (redact before sharing)	_____
Command or test used	_____
Result / what it means	_____
Issue noticed and safe next step	_____

Final report template

Title: Cybersecurity Fundamentals and Local Network Observation

1. Objective
Write two or three sentences explaining what you intended to learn: cybersecurity fundamentals, basic network configuration and safe observation.

2. Environment and authorization
Record your device, operating system, lab type and confirmation that you own or are authorized to use the system. Do not disclose sensitive identifiers.

3. Concepts learned
Summarize the CIA Triad, at least three threats, attack surface, IP versus MAC, ports, OSI/TCP-IP models, DNS, DHCP, HTTP/HTTPS and authentication versus authorization.

4. Procedure
List the commands or settings pages used, in the order performed. Explain that they were used only to inspect your own device or isolated lab.

5. Observations and results
Include a redacted table of observations. Explain what the IP configuration, gateway, DNS server and local listening sockets mean. Add screenshots only after removing personal or sensitive details.

6. Security recommendations
List practical improvements such as updates, MFA, backups, least privilege, secure Wi-Fi and disabling unused services.

7. Conclusion
State what you learned, what worked, any limitation and one topic you want to study next.

Submission checklist

- Learning notes are complete and organized by day.
- Practical record contains date, environment, commands and observations.
- All activities were performed on an owned or authorized system.
- Screenshots and outputs have been checked for passwords, tokens and personal information.
- Report includes objective, procedure, results, recommendations and conclusion.
- LinkedIn post (if required) describes learning without exposing sensitive network details.

Sample LinkedIn post

I completed Module 01: Cybersecurity & Networking Foundations (Day 1–Day 5). I learned the CIA Triad, common cyber threats and attack surfaces, networking basics such as IP addresses, MAC addresses and ports, and the OSI and TCP/IP models. I also explored HTTP/HTTPS, DNS, DHCP, authentication and authorization. As a practical exercise, I reviewed network configuration on my own device in a controlled environment and documented the observations. This module strengthened my understanding of safe cybersecurity practices, including MFA, updates, least privilege and responsible testing. #Cybersecurity #Networking #Learning

Revision questions and short answers

Use these questions to test your understanding. Expand each answer with a labelled diagram or example when writing a longer university-style response.

1. Define cybersecurity and explain the CIA Triad.

Cybersecurity protects systems, networks and data against unauthorized access, misuse, disruption or damage. Confidentiality restricts access to authorized parties; integrity protects accuracy and authorized changes; availability ensures services and information are accessible when needed. Examples include encryption, hashes and backups.

2. Differentiate threat, vulnerability and risk.

A threat is a potential cause of harm. A vulnerability is a weakness that may be exploited. Risk considers the likelihood and impact of a threat exploiting a vulnerability. Patching reduces vulnerabilities and therefore can reduce risk.

3. What is an attack surface? How can it be reduced?

The attack surface is the set of possible entry or impact points in a system, including accounts, applications, network services, devices and people. Reduce it by disabling unused services, patching, MFA, least privilege and limiting exposed interfaces.

4. Differentiate IP and MAC addresses.

An IP address is a logical network-layer address used for routing between networks. A MAC address is a link-layer identifier used for delivery on a local network. IP addresses can change with network configuration; MAC addresses may also be randomized or changed.

5. Explain the OSI model.

The seven layers are Physical, Data Link, Network, Transport, Session, Presentation and Application. They describe functions from physical transmission to application services and help organize network troubleshooting.

6. Compare TCP and UDP.

TCP is connection-oriented and provides reliable, ordered delivery with retransmission and flow/congestion control. UDP is connectionless and has less protocol overhead but does not guarantee delivery or ordering. Applications choose based on their requirements.

7. Explain HTTP and HTTPS.

HTTP transfers web resources. HTTPS is HTTP protected by TLS, which provides encryption in transit, integrity protection and server authentication. Users must still verify the domain and be alert to phishing.

8. Explain DNS and DHCP.

DNS resolves domain names to records such as IP addresses. DHCP automatically provides network configuration such as IP address, prefix, gateway and DNS server. The common DHCPv4 sequence is Discover, Offer, Request and Acknowledge.

9. Differentiate authentication and authorization.

Authentication verifies identity, such as by password plus MFA. Authorization determines what an authenticated identity may access or do, such as viewing but not editing a record.

10. List the steps for creating a safe cybersecurity lab.

Use owned or explicitly authorized devices, isolate virtual machines, use host-only/internal networking when possible, take snapshots, use sample data, keep systems updated and do not test public or third-party targets without permission.

11. What is a port number?

A port number identifies a service or application endpoint on a host for TCP or UDP communication. Examples include TCP 443 for HTTPS and TCP 22 for SSH. A familiar port number does not prove that a service is active.

12. Why should practical findings be documented?

Documentation makes work repeatable and understandable. It records the environment, authorization, commands, observations, results, limitations and recommendations, while allowing sensitive information to be redacted.

Glossary

Access control	Rules determining who or what may access a resource.
API	Application Programming Interface; a defined way for software components to communicate.
DHCP	Protocol that automatically supplies network configuration.
DNS	System that resolves domain names into records.
Encryption	Transforming data so it cannot be read without the appropriate key.
Firewall	A control that allows or blocks network traffic according to rules.
MFA	Multi-factor authentication; use of multiple different factor categories.
Packet	A formatted unit of data carried by a network-layer protocol.
Router	Device that forwards packets between networks.
TLS	Transport Layer Security; protocol used to protect communications.
Vulnerability	A weakness that could be exploited.
Zero trust	Security approach requiring explicit verification rather than implicit trust.

End of Module 01 notes. Recommended next step: complete the practical record on your own device, then write the final report using the template above.

Reference note: These notes summarize widely used networking and cybersecurity concepts for beginner education. For formal study, consult your course material and current official documentation for your operating system and network tools.